



温州肯恩大学

WENZHOU-KEAN UNIVERSITY

MGS 3001 WHS01
Python Programming for Business

Midterm Exam Feedback + AI Fluency Cont.

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Week 7-2, 11 April 2026

Midterm Exam Feedback

Class average: **23.06 / 30**

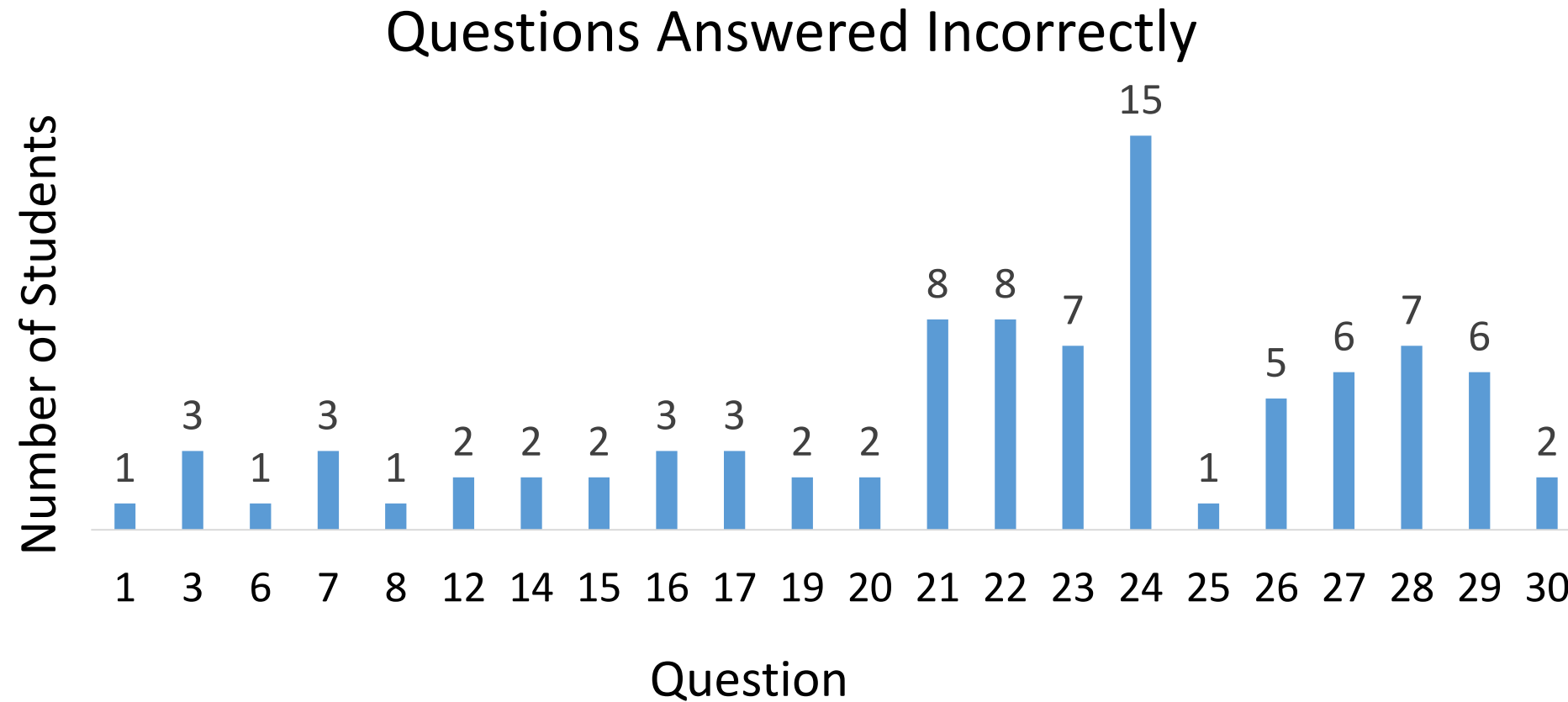
- Grades have been updated in Canvas.
- If you have any questions about your result, or if you would like to review your midterm exam paper, please come to my **office hours**.

Tue & Thu 11:15 – 12:30 | Wed 14:00 – 16:30

Please note:

- **Do not worry too much** if your score was below your expectation.
- There will be **opportunities for bonus** points later in the course.
- Final grades (A, A⁻, B⁺, B, etc.) will be determined based on **my overall grading rules** at the end of the semester.

Midterm Exam Feedback



Midterm Exam Feedback

24. Which function calls are valid for the definition below?

```
def order_summary(price, quantity, discount_rate=0.1):  
    return price * quantity * (1 - discount_rate)
```

- A. `order_summary(100, 2)`
- B. `order_summary(quantity=2, price=100)`
- C. `order_summary(price=100, quantity=2, 0.2)`
- D. `order_summary(100, 2, discount rate=0.2)`

Midterm Exam Feedback

21. Which of the following names are valid and appropriate in Python for variables?

- A. `total_sales`
- B. `2nd_customer`
- C. `calculateRevenue`
- D. `class`

Rules for Python variable name:

- Must start with a letter or the underscore character
- Cannot start with a number
- Can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Case-sensitive (age, Age and AGE are three different variables)
- Cannot be any of the Python keywords (e.g., `def`, `class`, `True`, etc.)

Midterm Exam Feedback

22. Which of the following statements about Python functions are correct?

- A. A function can have default parameter values that are used when no argument is provided.
- B. A function can be called using keyword arguments, such as `f(x=3)`.
- C. A function must always include a return statement; otherwise, it will cause an error.
- D. The order of arguments matters when calling a function using positional arguments.

Midterm Exam Feedback

23. Which of the following statements about Python control flow are correct?

- A. In an `if-elif-else` structure, once a condition is true, the remaining conditions are not evaluated.
- B. A `for` loop always executes at least once, even if the sequence is empty.
- C. A `while` loop continues executing as long as its condition remains true.
- D. The `else` block in an `if-elif-else` structure runs only if all previous conditions are false.

Midterm Exam Feedback

26. Which situations can raise runtime exceptions?

- A. `10 / 0`
- B. `numbers[100]` when the list is too short
- C. using an undefined variable name
- D. missing a colon after `if`

Midterm Exam Feedback

27. Which statements about tracebacks and debugging are correct?

- A. The last line of a traceback usually shows the exception type and message.
- B. Printing intermediate values can help isolate a logical error.
- C. A logical error always produces a traceback.
- D. Reading the traceback can help locate the failing line.

Midterm Exam Feedback

28. Which statements about Python program execution are correct?

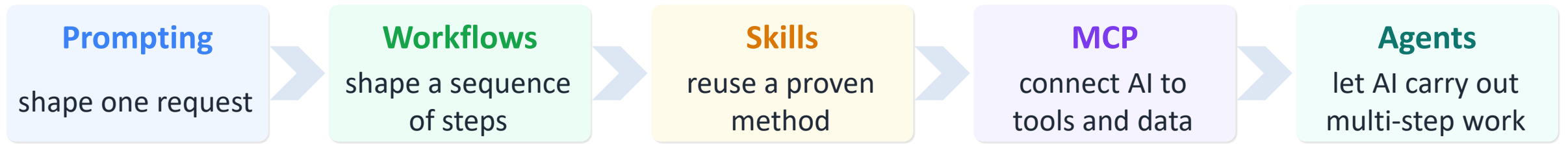
- A. Python executes code from top to bottom unless control flow changes the order.
- B. Code inside a function definition runs immediately when Python reads the function definition.
- C. Code inside a function runs only when the function is called.
- D. Python ignores indentation when determining program structure.

AI Fluency cont.



From Prompts to Agents

These concepts all build on the same core interaction: **giving AI instructions and context to make outputs more useful and reliable**. What changes is the scope being structured — one request, a repeatable method, a workflow, or a larger work environment.



Two newer ideas sit across that ladder:

Context engineering Decide what information, instructions, and tools the model should see at each step.

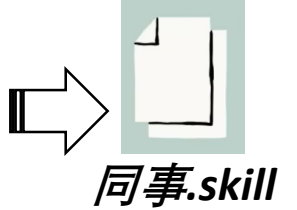
Harness engineering Design the broader environment around the model: memory, tools, checks, feedback loops, and human review.

Takeaway: do not chase buzzwords. Ask which level of structure your task actually needs.

From Prompts to Agents

Agent Skills with Anthropic

蒸馏同事



Prompting

One request, one response

Best for quick drafting, summarizing, brainstorming, or clarifying one task.

Example: "Summarize this report in 5 bullets."

Workflows

A sequence of steps

Best when the work includes multiple stages, handoffs, or revisions.

Example: collect data → analyze → draft → human review.

Skills

A reusable method

Best for recurring tasks where you want consistency across many conversations.

Example: a standard method for analyzing customer complaints.

A prompt is one question. A workflow is a process. A skill is a reusable recipe inside that process.

From Prompts to Agents

When AI needs external systems and multi-step action

MCP Model Context Protocol

A standard way for AI applications to connect to files, databases, APIs, search tools, calendars, and other systems.

files

database

search

calendar

Agent AI configured for multi-step work

An agent can plan, use tools, check results, and continue in a loop — instead of answering only once.

Plan



Use tool



Check result



Continue / stop

Without connections, AI mostly talks. With MCP and agent design, AI can access systems and carry out longer tasks more usefully.

From Prompts to Agents

Two broader ideas for making AI *more reliable over longer work*

	Context Engineering	Harness Engineering
核心关注	喂给模型什么信息	Agent整个运行环境设计
解决的问题	如何在有限的上下文窗口里，动态组装出最相关的“信息包”	如何构建一套系统，让Agent在长时间、复杂任务重不出错、不跑偏
工作范畴	信息选择、组织、格式化	包含约束、反馈、验证、状态管理等在内的完整系统设计
比喻	给马提供一张详细的地图和路标	为马搭建一套包含缰绳、马鞍、围栏和赛道的完整马具系统

Limitations of Generative AI

All “story” begin here

Limitations of Generative AI



Knowledge
cutoff date



Hallucination



AI context
window



Non-deterministic
output

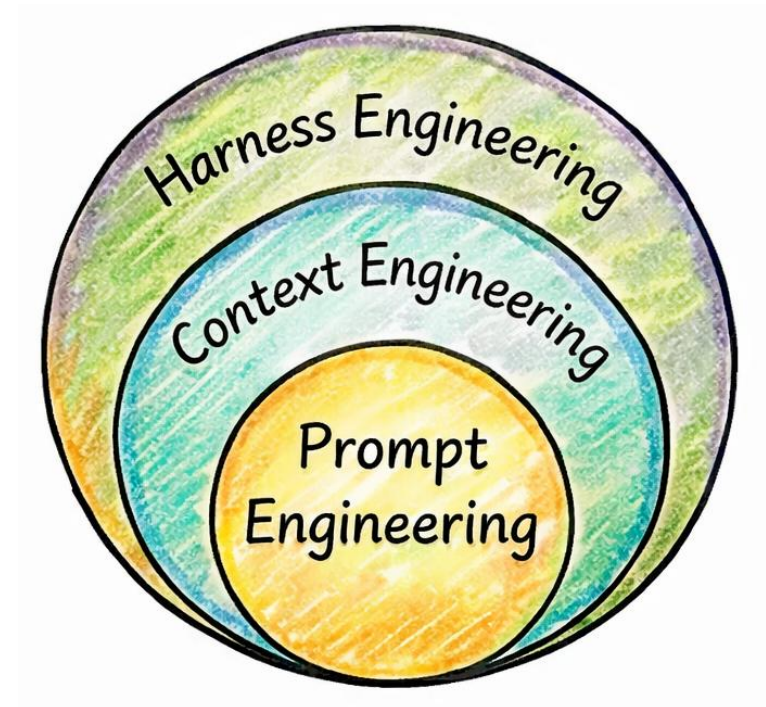
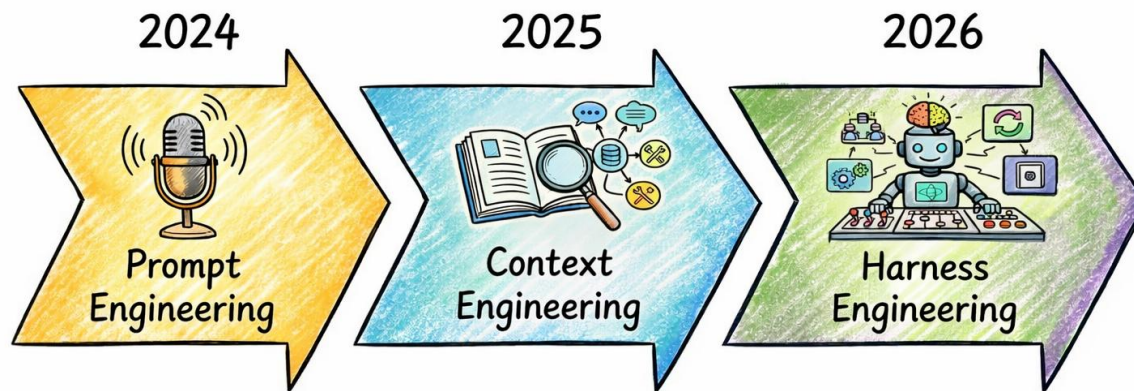
抽卡



Reasoning
limitation

From Prompts to Agents

AI工程范式转变



Generated by ChatGPT Thinking

From Prompts to Agents

Use the ideas, but do not get trapped by the buzzwords

- Fundamentally, these ideas all build on the same core interaction: giving AI instructions and context to improve output **quality** and **reliability**.
- Most of the newer concepts exist because current generative AI still has practical limits — inconsistent outputs, limited context, and difficulty handling longer workflows reliably.
- As models and user interfaces improve, some of these methods will become less visible or more automated for non-technical users.
- You do not need to master every term. You need to find a way of working with AI that fits your task, your standards, and your professional context.

The toolset will keep changing. Your judgment about how to work with AI is the more durable capability.



After-Class Exercise

- Register for a GitHub account if you do not already have one.
- Install Git on your computer; GitHub Desktop is optional.
- Create a GitHub repository for this course – MGS 3001 Python Programming for Business
- Upload the Jupyter Notebook you created in Week 6-2 to your repository
- Explore how Git, GitHub, and VS Code work together:
 - write, run, and edit code in VS Code
 - manage changes with Git
 - update your repository on GitHub for version control
- Explore useful open-source projects on GitHub, especially those related to:
 - AI applications
 - Python scraping websites or platforms relevant to your research project data collection

Q&A

CBPM B214

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